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Energy and delay efficient fog computing using caching mechanism

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ABSTRACT

Fog computing has emerged as an extension to the existing cloud infrastructure for providing latency-aware and highly scalable services to geographically distributed end devices. The addition of the fog layer in the cloud computing paradigm helps to improve the quality of service (QoS) in time-critical and delay-sensitive applications. Due to the continuous increase in the deployment of fog networks at large scale, energy efficiency is a significant issue in the fog computing paradigm to reduce the service cost and to protect the environment. A plethora of research has been conducted to reduce energy consumption in fog computing, majorly, focusing on the scheduling of incoming jobs to improve energy efficiency. However, node-level mechanisms have largely been neglected. Cache placement is a critical issue in fog networks for efficient content distribution to clients, which requires simultaneous consideration of many factors including quality of network connection, the demand for contents, and users' activities. In this paper, a popularity-based caching mechanism in content delivery fog networks is proposed. In this context, two energy-aware mechanisms, i.e., content filtration and load balancing, have been applied. In the proposed approach, popular contents are found using random distribution and these contents are categorized into three classes. After finding the file popularity, an active fog node is selected based on the number of neighbors, energy level, and operational power. Further, the popular content is cached on the active node using a filtration mechanism. Moreover, a load-balancing algorithm is proposed to increase the overall system efficiency in the cached fog network. The evaluation of the proposed approach exhibits promising results in terms of energy consumption and latency. The proposed scheme consumes 92.6% and 82.7% less energy in comparison to *without caching* and *simple caching* mechanisms, respectively. Similarly, an improvement of 85.29% and 67.4% in delay has also been noticed while using advance caching against the *without caching* and *simple caching* techniques, respectively.

1. Introduction

Smart cities encompass a vast array of IoT devices providing various services through a massive amount of data captured from the surrounding environment. This data is extensively analyzed for improving quality of life and enabling various smart city services. End-user devices though capable yet are not equipped with the required computing power for processing this considerable amount of data. Similarly, these end-user devices require reliable and real-time access to content, encouraging service providers to employ cloud computing for improving the overall process. In this context, cloud computing has emerged as the de-facto storage and processing paradigm to provide the user's required services for their computational and storage needs. However, cloud computing possesses several underlying networking issues due to the centralized nature and remote storage of resources. Hence, it is not considered an ideal choice for the time-critical application. To address these issues, CISCO proposed a novel paradigm named Fog Computing.

Fog enabled networks bring the storage and processing capabilities to the near vicinity of end devices [1]. Fog nodes are connected to the cloud where the services request are ultimately transferred in case the Fog nodes are unable to complete the requests. Several topology configurations have been proposed where various configurations are attained in order to effectively respond to the end-user requests based on near vicinity of the edge nodes. In order to better facilitate

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the end-users requests, the Fog nodes share their work load by corresponding with each other and sharing work load.

Researchers have attempted to utilize various techniques for processing load balancing [2]. Similarly for provision of storage based serves Caching is an efficient solution to increase the energy efficiency and performance. In [3], caching is used to reduce energy efficiency, but their work has issues like load balancing and energy hole creation. Authors have chosen a mechanism where they are selecting the top node based upon availability of space, and all of the future requests go to the top node in the list this creates an energy hole problem in fog load balancing.

The filtration module along with the cache is introduced to answer one of the primary research questions. To resolve this issue, a filtration module has been added, which ensures only those files that are replicated on such a node that can process them. Apart from this, it is also proposed to leverage the load balancing algorithm to cater for the efficient balancing of load among the participating nodes.

Another problem is that if the user request miss-match with the top node computational capabilities, then caching does not perform well. So based on these shortcomings, in this paper, an advanced caching method, i.e., a caching with filtration mechanism and load balancing technique, is proposed that removes the issues highlighted above.

1.1. Motivation

Fog computing is a new field, with great potential for enabling future technologies, and lots of work needs to be done in energy efficiency schemes. Fog computing layer is an energy-efficient solution of the cloud computation network as it reduces the energy of the cloud layer server and reduces the latency because it is close to the network edge. The focus of this research is to make the fog network energy efficient as it is an emerging future technology, and if the fog network is made energy efficient, then it will also increase the overall system performance. In fog network, several gaps are present like resource scheduling, load balancing, security, latency, and energy, and more will be explored in the future. There is a need to address these gaps for making this new paradigm up-to the standards. However, in this work, an in-depth analysis is performed related to energy efficiency mechanism for fog network. Furthermore, from an extensive literature review and comparative analysis of the previous work, several new challenges have been deduced, and in the text to follow, it is attempted to explore new opportunities for the research.

1.2. Background

Cloud Computing: Due to limited constrains of IoT, many real-time applications (such as health care) require high bandwidth and processing power. Managing and storing huge amounts of data locally and temporarily will not be possible. For this purpose, external rental storage space is required. The data is not only formed to information, and to knowledge, but it should also be made a means of wisdom for the user. Such type of processing is not possible at IoT devices. Therefore, cloud is connected with IoT devices to overcome theses limitations and to perform data analysis efficiently with a large scale backup of computational and storage devices. Furthermore, cloud computing extends the traditional cloud services to is sent to the edge of IoT devices. Moreover, the data generated from IoT devices is forwarded to the cloud for further operations and storage. The result of the processed data is acknowledged back as per user requirements [4]. As the number of requests to the cloud is gradually increasing, it creates congestion on the network due to the centralization of the resources. Due to these reasons, the cloud network is not an ideal candidate for time-critical applications. To cater with above-mentioned issues, the researchers propose a novel paradigm called fog computing [5].

Fog Computing: CISCO proposed a new paradigm to provide network support for real-time, low latency, mobility, and location-aware

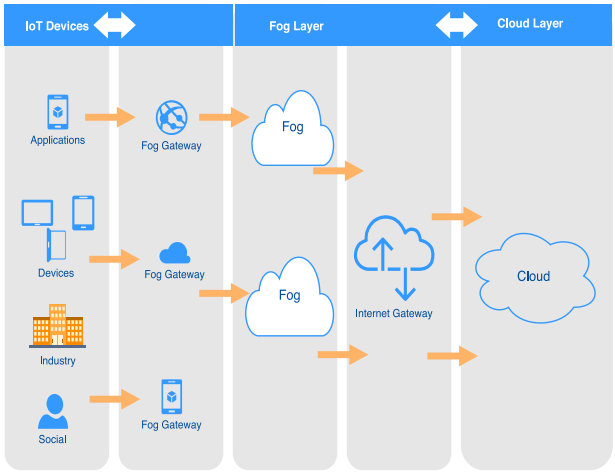


Fig. 1. Network model.

services. A huge amount of IoT devices are located on the network edge need energy-efficient, real-time, and time-critical services. For this purpose, multiple fog devices, with computational power to process the data, are available at different locations in the network. Hence, fog network is introduced to use its computational power and resources to perform some of the cloud services on the network edge.

Fog network is an intermediate layer between IoT and cloud, also called a mini cloud. It is introduced to extend the cloud computing near to the data generation sources. Fog network decides intelligently that which data would be processed on the fog layer and which request to be forwarded to the cloud computing services.

Fog computing is not an alternative solution to cloud computing networks, but it is introduced to complement the cloud network. The main goal of these fog and cloud layers is to provide different user services that they cannot perform on their devices. Fog/cloud computing is used to - serve the user requirement, support complex operations, and long term data storage at the network core (see Fig. 1).

The paper is organized as follows. The problem statement is presented in Section 1, Section 1 is the introduction. Section 2 has related work. Section 3 introduces our proposed advance caching approach. Experimentation and evaluation are performed in Section 4. The paper end with the concluding remarks in Section 5.

2. Related work

In [6], they focus on problems related to energy consumption of developing clusters. They propose a fog computing-based energy-aware load balancing and scheduling (ELBS). Related to the workload on the fog node, they develop energy consumption and optimization function for the cluster load balancing. Particle Swarm Optimization (PSO) is used for achieving the task and improving performance. Finally, to achieve the distributed scheduling of clusters, they introduce a multi-agent system.

In [7], they worked on balancing of the energy consumption and services delay in homogeneous fog network and developed the cross-layered analytic framework. They have developed a novel approach Delay Energy balanced tasking scheduling called DEBTS using Lyapunov optimization for reducing energy consumption. They also prove DEBTS performance using theoretical modeling. They also developed an online algorithm for handling stochastic modeling of the system as in its prior information related to fog nodes is required. Their performance matrices are energy consumed during task scheduling and task execution.

In [8], Fog Node has a limited amount of energy available because most of them are renewable energy sources powered, so efficient energy management for computing and communication is required in

fog nodes. The main goal of their work is to find the application which consumes more energy on the fog node. They have proposed a penalty-aware bin packing optimization problem for the dynamic resources management of the transport throughput, container processing in TCP/IP based Fog DCs. The heuristic works by adaptive increasing and decreasing the processing speeds of VMs using hard bounds.

In [9], they propose for task admission an asymptotically optimal online programming for reducing energy consumption and also decreasing the time complexity. Their technique also ensures task delays. In their approach, the task which failed to meet deadlines is selected for offloading. It makes this problem Integer programming with optimal substructure property. They also discretize energy saving. Using Dynamic programming, they developed an energy-efficient offloading and resources optimization scheme (EROS) to minimize the energy consumption for delay-sensitive tasks.

In [10], they developed an analytical model to solve the energy-efficient offloading in a homogeneous fog network. They also proposed a mathematical based algorithm named as maximal energy-efficient task scheduling (MEETS) for optimal task node scheduling.

In [3], they use a caching technique to optimize the performance of the fog network. Using caching placement, they propose that energy efficiency can be achieved. In their algorithm, using Zipf distribution, they classify files into three categories (high, medium, and low) based on popularity. After file classification, they select the popular node and replicate the file on the high-level fog nodes. Selecting the node in the cluster head for caching the file is also their key contribution in this paper. They have tested their algorithm using energy efficiency simulations.

In [11], they worked on resource allocation problems for the mobile fog network and proposed a green fog computing solution to increase energy efficiency. Their problem falls into NP-Hard because it is mixed-integer nonlinear programming (MINLP). MINLP increases energy efficiency, and it works in distributed form. To solve the MINLP problem, they propose their novel algorithm NOVA which uses the Dinkelback algorithm and benders decomposition. They have simulated their proposed algorithm in Matlab with MOSEK solver. They have used maximum transmit power and total power consumed as the performance matrix.

In [12], Fog servers are not as powerful and energy resourceful as cloud servers, thus, a data processing balancing is required between the cloud and fog servers. Energy efficiency for the fog nodes is among key concerns for the researcher, and this paper also addresses this issue. Green energy resources are unreliable, unpredicted, and dependent on the environmental condition, therefore, a mechanism is needed that makes sure the energy availability for fog servers. They propose a dual power source for operating fog nodes and also developed an online algorithm using the Lyapunov optimization technique on time and energy and named it LOTEC. The main goal of this algorithm is to make sure the data is processed within time limit fog server availability is guaranteed, keeping in mind that overall system cost is minimized.

Fog nodes have multiple power sources like renewable energy sources and the main electric grid. As renewable energy resources are unpredicted and unreliable, and it is dependent on environmental factors like wind or solar energy. Thus, there is a need to maintain the backup to make sure the operations are performed without any break. In this paper, the author works on optimizing the energy management in fog network and in time of peak works system cope with it even if power generation is low. In [13], using the reinforcement learning approach, they deal with the process of job arrival and jobless probability. Markov-based analytical modeling is performed along with the reinforcement learning approach for server activation optimization.

Traditional cloud computing faces issues like high service latency, lack of location-awareness, and user mobility support. Compared to cloud computing, fog computing can address these problems by utilizing elastic resources of fog nodes for providing the enhanced service quality with low latency. However, there are still lots of energy constraints with the fog nodes. In [14], they proposed a novel algorithm called Cachinmobile for the energy-efficient edge node using a device to devise communication and social networking. D2D is used to communicate between mobile devices directly in short distances without using wireless infrastructure.

In [15], they studied Fog network and their characteristics over Fog DCs. They also gave a holistic fog-supported architecture that manages the networking cost and computation in a variable network traffic case. The main feature of their work is real-time communication using fog layer, FDC based IOT environment, and its characteristics. In [16], the authors work on task scheduling and selection at the edge nodes. They proposed their own energy-efficient task scheduling and selection algorithm called container-based edge service management system (CoESMS). CoESMS used cooperative game theory. They are using popular Container as a Service (CoaaS) for improving the fog node's scalability and performance. Cooperative Games strategy is used for the scheduling and selection of the task. In [17], the author works on the adaptive energy-efficient orchestration and reconfiguration of the virtualized communication and computing.

In [18], "Nano Data centers" known as small servers, these are used for content delivery, hosting, and P2P application are in end-user premises. In this paper, they perform the comparison of cloud computing and fog computing based on centralized data centers and Nano data centers. They also aim to find the cases in which applications perform well on nano data centers with respect to energy efficiency. They used a measurement-based model to calculate the network energy consumption. In their experimentation, they found that energy is saved using nano data centers. In [19], Energy efficiency is an important issue in ad-hoc networks like wireless sensor networks. In WSN sensor have resource constraint, so it is more important to conserve the resources because it affects the network lifespan.

Energy optimized stable cluster head selection is the current key challenge in the WSN domain. In this paper, they worked on this issue and proposed a new protocol for node clustering, named a new stable election protocol (NSEP). NSEP maintains energy consumption and increases the lifespan of the sensor network for a stable period. Their proposed scheme works in this way that clustering structure is used for routing between the fog node, also called a sink node, and sensor node to prolong sensor network lifespan in an energy-efficient way.

In [20], offloading the computationally expensive tasks to the network edge, fog computing technique is currently used. Massive bandwidth and resources are required because a huge amount of data is transferred between mobile devices and fog servers. For improving the efficient utilization of bandwidth and resources, a new approach is proposed in this paper. Non-orthogonal multiple access (NOMA) used for hybrid computation offloading, user have independent though multiple tasks which can be processed at fog nodes or on cloud servers.

In [21], the author works on Body sensor networks. In healthcare, now Body sensor network (BSN) is used. These BSNs are battery enabled, so to make them energy efficient is a key goal because it increases the sensor life. In this paper, using fog computing concepts in the e-health field, they proposed energy efficiency in BSNs. In their work, they perform the comparison of Fog based BSNs and traditional BSNs to test their proposed approach and show its performance. In their work, Fog computing on a gateway was introduced to add on the performance of the health monitoring system. They also implemented different fog computing services like ECG feature extraction, a graphical interface with access management, distributed database and push notification to achieve energy efficiency at a sensor node and reducing the load in a cloud server.

Lots of work has been done in the fog computing field, but modeling of fog networking was untouched area, so in [22], focus on theoretical modeling of fog computing architecture. Performance comparison of cloud and fog computing is also performed in this paper. Fog computing is an efficient green computing platform for future IoT applications than cloud computing. Fog computing consumes 40.48% less energy than

cloud computing. Their performance metrics include service latency and energy consumption. They have discussed the few real-life systems in which fog computing helps like Smart vehicle management and smart grids.

In [23], a Huge amount of data is generated from IoT clusters and efficient data processing required. In this research paper, they work on challenges related to architecture. They proposed a novel integrated fog cloud IoT (IFCIoT). The IFCIoT architecture aims to increased scalability, quick speed time, performance, energy efficiency, and better-localized accuracy for IoT/CPS applications. In [24], the author proposed a new framework to integrate microgrid and edge computing. They make sure that integration between these two complement others and enhance resource utilization. In their scheme, stats related to power generation and consumption are shared. As stats are shared, they help in timely decision making related to energy consumption. Microgrid supplies the required energy and estimates the future requirement of the energy that the grid must supply to ensure the smooth running for the edge computing network. As continuous energy is required for smooth processing of task, their proposed framework works for this and makes sure when a critical task in the running, energy is available all the time.

In [25], they introduced the task offloading network architecture for multiple fog nodes. Performance is based on energy consumption and task delay. Fair Task Offloading (FTO) scheme is proposed in two steps. First, they select the fog nodes based on the probability value that comes from the FTO metric. In the second step, they offload the task to the selected node using the above metric result and rules of FTO to minimize the task delay and increase the performance. Their testing results show that their scheme has better results and improve energy consumption and decrease the processing delay.

2.1. Comparison analysis

A comparison with the past research paper has been performed such that their evaluation strategy, techniques, approach, tools, language, and their major contributions were explored and the comparison were performed on these metrics.

From Table 1, it can be deduced that in the past research, they have mostly worked on the analytical modeling of the problem, and using simulations, they evaluate their approaches. It shows that the practical deployment or real-life applications of the work is not present at the moment. It also shows that things may differ or results may change when it is actually applied in the heterogeneous network, as most of the works have considered the homogeneous fog network.

From the comparison of the techniques, most of the problems falls into the stochastically category, and their solutions are NP-Hard or Complete, which shows that improvement can be made in the modeling through which the NP-hard problem can be converted into P-type problem that may help in achieving more efficiency. Most of them have used a simulator to simulate their algorithms. Their training set is also small, which may cause performance degradation when their approaches are tested in real life or on a large fog network.

3. Methodology

Based on the entire research survey and comparison metrics, some issues were also found in the research work carried in [3] and in their proposed algorithm. In [3], which is the popularity based caching method, the authors are selecting the popular file using Zipf distribution. After file classification as popular, medium popular and less popular, they replicate the file into the active node. They are selecting an active node on the basis of most number of edges or connection with other nodes. Then request processes from the most active nodes.

But there are few issues associated in their approach like popular file are copied on the node which may not have the capability to process that information, so there should be a filter before replicating file that ensures that popular file gets copied to only those active nodes which have the computational capability to process it. Secondly, when the active node is selected, then request are forwarded to active node, which drains that node power, and it will go to dead state, and again system have to find the new active nodes which add on to the performance and increases the overall system delay and energy consumption. So load balancing is also required for resolving this issue.

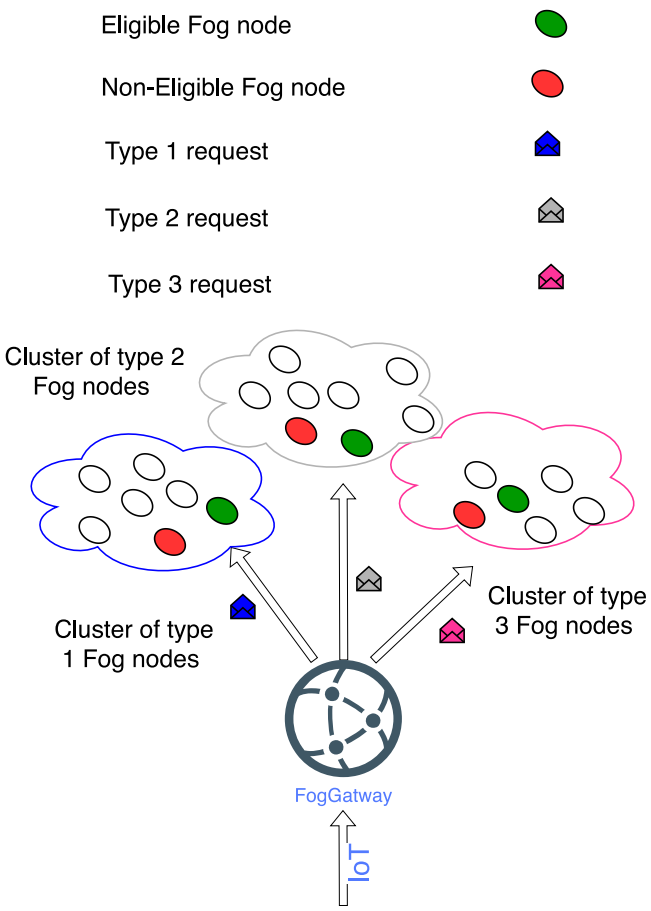


Fig. 2. Proposed system model.

Based on the highlighted above issues, an algorithm has been proposed that may resolve these issues. For the filtering mechanism, an extra layer is added after file popularity selection, which will make sure that only those files will be copied to the nodes which will have the processing power. And for the load balancing, the decision of the node is made based on computational node capacity, computation availability, active nodes, and current energy level. Using these four parameters, the proposed load balancing algorithm works and selects the node for processing the request, which will have a high energy state level and high computation availability. The proposed model is based on the system model, as shown in Fig. 2.

The caching technique is utilized to achieve the energy efficiency. Along with caching, a load balancing algorithm is also introduced to avoid the problem highlighted in [3]. The caching process is repeated every 30 min and the beacon packet is sent that contains only the node information which is selected to be at the top level. Fig. 3 shows the format of the beacon packet.

Using Eq. (1), the average response time of the simple cache-based fog network is calculated without the load balancing method, which is used as a benchmark. In Eq. (1), N means the total number of fog nodes in the fog layer network, s is the number of searches performed, and L1 is the network delay, which is dependent on the Internet.

$$RT_s = N * s + L1 \tag{1}$$

Table 1
Comparison matrix. In the evaluation column, P means Prototype, E means Experimentation, S means Simulation, X means No evaluation. In approach column, MM mean Mathematical Modeling, AF mean Analytical Framework, H means Heuristic, S means Scheme, AM means Analytical Modeling.

Name	Evaluation	Technique	Performance metric	Major contribution	Approach	Tool
Wan et al [6]	P	CSCS method, ELBS method	PSO Algorithm, Multiagents, and Candy packaging line.	ELBS method proposed based on Fog nodes. • Multi-Agent system for dynamic scheduling for Job shop problem • Mathematical Modeling of load balancing using PSO Algorithm	MM	Raspberry pie board, UDOO board, ESP8266
Yang et al. [7]	S	Averaged delay, Energy consumption	Lyapunov optimization technique	DEBTS algorithm for minimizing the overall energy consumption while reducing service delays and Jitters delays. • Cross-layer analytical framework for balancing energy consumption and service delays.	AF	X
Naranjo et al [8]	S	MBFD and MDC, Microsoft RAID, Resource manager	Container based virtualization	PABP is used for the dynamic resource management heuristic. • Reduction of the overall energy consumed.	H	ifogsim
Lyu et al. [9]	S	B&B, ARAA, and Local	Dynamic programming	EROS is designed to minimize the total energy consumption of devices under latency constraints.	S	Monete Carlo
Yang et al. [10]	S	ETO, EETO, and AMC	Rigorous mathematical analysis	MEETS for energy-efficient task scheduling in homogeneous nodes.	AM	Face recognition application
Althamary et al [3]	S	GMAC-MT,PAC-MT, PAC-UT	Zipf distribution	Caching placement in fog nodes to reduce energy utilization.	A	Matlab, C++
Yu et al [11]	S	Utility function, Total Power, and Power consumed.	Bender decomposition, Dinkelback algorithm, and modified NOVA	Maximizing the utility function w.r.t energy efficiency for the mobile fog network.	A	Matlab with MOSAK solver
Nan et al [12]	S	Waiting queue in CDC, Avg Response Time, Avg Cost. Fog-Only, Cloud-Only, and Round Robin algorithms as our benchmarks	Lyapunov optimization, Stochastic optimization, and online algorithm.	An online algorithm Lyapunov optimization on time and energy cost for balancing data processing time and system running cost.	AF	DSC using simpy
Conti et al [13]	E	Time job spends in node, Size of renewable energy generation, Avg delay in task queuing.	Markov based analytical model, Reinforcement Learning	Markov based reinforcement learning method to optimize the server activation policy. • Minimization of job loss probability through BESS.	AM	Matlab
Wang et al [14]	S	No caching, Random caching	Genetic algorithm, Zipf distribution	Addressed the content placement problem through an optimal caching strategy to minimize energy consumption.	AM	Zipf distribution

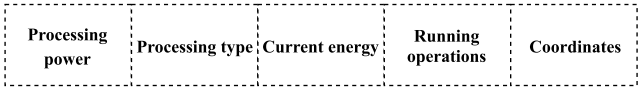


Fig. 3. Format of the beacon packet.

Eq. (2) is used to formulate and find the response time of the advance caching algorithm, which is proposed in this study. Here, C shows the number of nodes in the cluster, s means several searches, and L1 is the average network delay. Here C contains the number of nodes which is selected based on the request. According to our algorithm, among all the nodes, only those node will be selected, which have the capability to process the request. so C < N.

$$RT_a = C * s + L1 \tag{2}$$

Eq. (3) is for calculating the average latency of requests without any caching. It is also used as a benchmark. In it, L1 is the average network delay between user and fog layer, and L2 is the network delay between the cloud and fog layer, and s is search time.

$$RT_n = s + L1 + L2 \tag{3}$$

Every search or operation will take 1 ms of time, and s will be calculated on the basis of Eq. (4). L1 is the network delay, which is set to 5 ms and L2 to 5 s N, C, and s are variable and depend on running of the stimulation. In Eq. (4), n is the number of nodes used in search space, R_op is currently running operation in ith node, and p is the cost of operation which is set to 1 ms.

$$s = n * R_o p * p \tag{4}$$

For energy calculation, Eqs. (4)–(6) have been utilized. In the energy calculation equation, u is the amount of energy consumed per task, and it is set to 1 unit.

$$E_S = \sum (N * R_o p * u) \tag{5}$$

$$E_A = \sum (C * R_o p * u) \tag{6}$$

The proposed algorithm is presented below where an assumption is made that the fog nodes are rechargeable and static (see Fig. 5).

3.1. Pseudo code for file caching with filtration

1. Compute file popularity using Random distribution
2. Categorized file into High, medium and less popular
3. Populate the sorted list of nodes based on the number of nodes linked with them.
4. Select the most active node and most popular file.
   a. If the most active node has operation capability to process most popular file then replicates on it else go to 4
   b. Copy the most popular file on the most active nodes.
5. Broadcast the list of the top-level node in the network
6. Repeat after required time

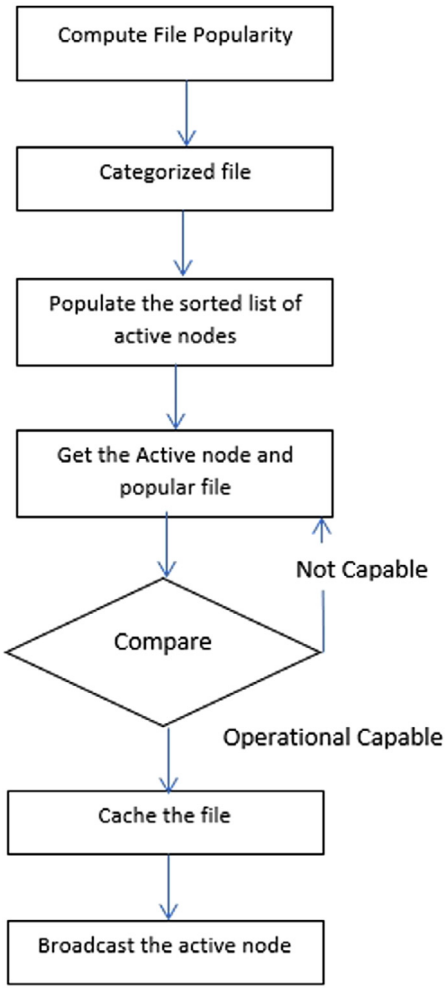


Fig. 4. Flow chart of Algorithm 1 (File Caching with filtration).

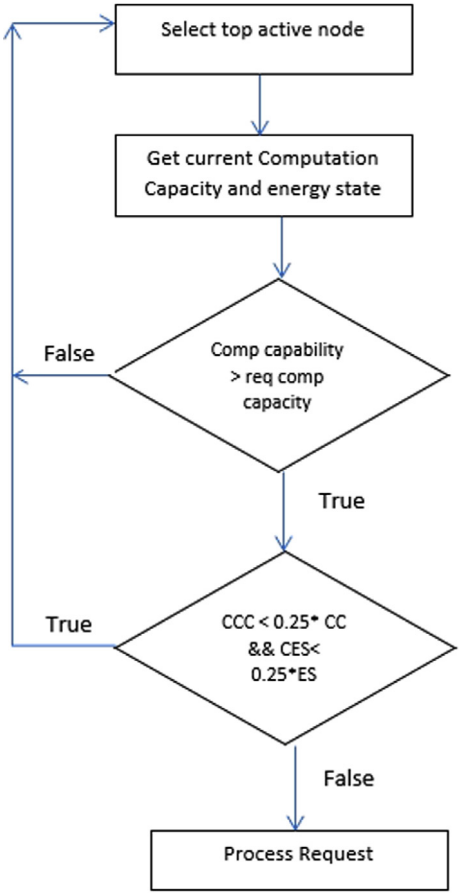


Fig. 5. Flow chart of Algorithm 2 (Load Balancing during Request).

3.2. Pseudo code for load balancing in request case

1. Select the top active node.
2. Get the current computation capacity and energy state of the node.
3. If (computational capability >required computation capacity) goto 4 else 1.
4. If (current computational capacity <0.25 * computational_ capability) OR (current energy_state <0.25*Total energy) then goto step 1 else step 5
5. Process the information.

4. Evaluation and experiments

4.1. Case study: Notice board

If you recall the purpose of the notice board, you will know it is made for spreading important information and saving the time of students instead of approaching the administration offices for getting the required information. In this context, three different scenarios have been considered to build up the proposed solution.

Scenario 1: No Notice Board : Considering the scenario of a university campus constituted of multiple departments where each department is situated in a separate building. Moreover, there is a building named administrative block where all the university administration including student affairs are settled. There are various information addressed to the students in a university. The university administration is majorly responsible to communicate the information to the students. Usually, this communications is taken place through the notice boards. If it is considered that there is no notice board in the building, then the students will go the administrative block to get any sort of required information. Hence, all the students of the university will rush to that block whenever there will be start of the semester, important announcements, results etc. Consequently, this building will be the main concentration for all the students that will result in a significant movement towards that building. As a result, the delay will be obvious in many situations. Similarly, some departments can be far away from administrative block and the students from those departments will get delayed to receive the information. In the view of distributed systems, this scenario can be mapped to the cloud where all the storage and processing resources are deployed and users from multiple locations have to approach it to fulfill their processing or storage requirements.

Scenario 2: One notice board : Now, the university set up a notice board in which they display announcements and important information. A student looks at the notice board and finds the information from there. If the information that they are looking at, they will go to the administrative office and get the information. In this case, a small number of queries will go to the administrative office while a large number of queries are addressed from the notice board, and it will save lots of time and decrease the burden from the administrative offices. But it raises the problem as there is one notice board, so everyone goes to one notice board, like a student from the third floor will go to the first floor for the information and vice versa. This is a good solution than the previous one, but it is not efficient because it increases the waiting time. If it is mapped to the network, it is an example of a fog network. Data is placed near edge users, and if the fog node does not

have the required information, then the gateway forwards the request to the cloud network. The problem, in this case, is load balancing.

Scenario 3: Multiple notice board : Currently, in CS building, every floor has a notice board and have the same information on all of them. In the previous case, one notice board was not an effective solution, so they placed a notice board on every floor and replicated the information on them. This is an efficient and time-saving solution, students do not have to go to another floor to get information, they will get it from the nearest notice board. Hence, this is what is proposed in computer networks that instead of centralized fog node, the caching technique should be used where the important data may be replicated, which can increase the overall performance.

4.2. Experimental setup

For the experimentation of the proposed algorithm, python is used as a programming language where the fog enabled caching simulation environment is created to evaluate the proposed algorithm. This environment provides all the underlying network functionalities and the protocols necessary to carry out a fog computing environment.

In the experiments, 10 fog nodes are used, and every fog node has 6 attributes - processing power, processing type, total energy, current energy, running operations, and coordinates. These attributed are used to design and develop the policies. Based on these attributes, caching and load balancing has been performed. The simulator works in a way that when the request comes, then it will first check the request type, after seeing the request type, it will find the related fog node which has the capability to process the node by scanning list of processing types from the list of the node. From the first scan, the subset of nodes is obtained, which has the capability to process the request based upon the demand. In the second pass, the current node's states are checked such that how much energy a node has and how much processing power is available. Thus, the simulator checks the node attribute "current energy" and "running Operations" to select the node. After this, the searching node is selected and the request is processed.

A user request to fog layer has four properties, which include - process type, required processing power, required energy to execute, and user location. Based on these properties, the proposed algorithm makes a decision to select a suitable node for forwarding the request. User location property is used to find the distance between fog node and user, it is used to allocate the fog node, which is screened out on the basis of other parameters. In addition, more than one node is used in the proposed work, and the fog node, which is located near the users, is selected for processing.

The values for calculating the latency and energy consumption are assumed, and using Eqs. (1)–(4), the simulations are performed and the response time is calculated on the proposed approach and two benchmarks, which are [3] and without caching. Eqs. (4)–(6) are used for calculating the energy consumption.

4.3. Results and discussion

The proposed scheme aims to improve the work conducted in [3]. Consequently, a comparative analysis has been performed between the proposed scheme and the base scheme [3]. The simulation is performed to evaluate the algorithm and compared it with the benchmarks. Fig. 3 shows the latency graph in which we have plotted the time taken to process the request on the Y-axis and no. of the file processed on the x-axis. This graph shows that our proposed approach has better latency than the benchmark. Initially, when the number of operations performed is less, in that case, the proposed algorithm has slightly high latency because of additional sorting that is performed to make a cluster of nodes for a request. However, on average, the proposed algorithm has better average latency than the caching based fog network and without caching fog layer network. The energy consumption graph also shows that when caching or without caching fog layer is used, then

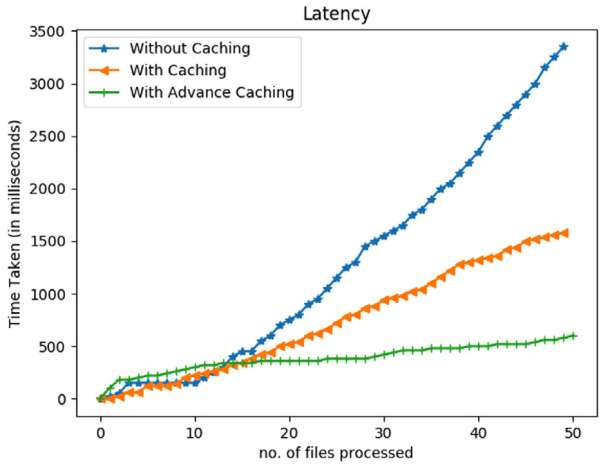


Fig. 6. Average latency.

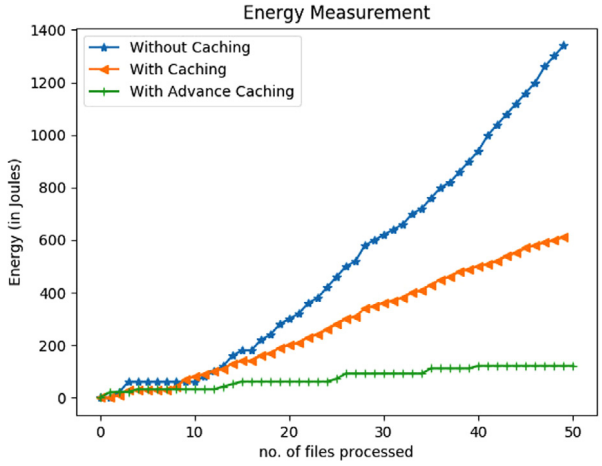


Fig. 7. Energy consumption.

fog nodes will drain out quickly and creates an energy hole. In Fig. 4, a graph of energy utilization is plotted to process the request. This graph shows how much energy is consumed when the same request is processed using simple caching, without caching and with advance caching fog layer network. In the graph, you can see that our proposed approach has consumed less energy than simple caching and without caching (see Figs. 6 and 7)

5. Conclusion and future work

Fog computing is an emerging research field that aims to provide an energy-efficient and optimal solution for the existing cloud network. The addition of the fog layer helps in critical and time-sensitive information to be processed in real-time, and it reduces the delay. Fog network also helps in improving energy efficiency. As this is a new network, so optimizing fog network in itself is an active research area. Lots of works have been performed related to improving the energy efficiency of fog network, but there is still room for more techniques to make fog network itself energy efficient. In this work, a detailed survey of the existing work is done related to energy efficiency in the fog layers. Further, an advance caching technique is proposed through which the energy efficiency and delays can be improved. Moreover, an algorithm for load balancing is also proposed in the advance cached fog layer. The caching is performed based on the capability attributes of the fog layer, which is not considered in the previous work. The obtained results show that the advance caching has 85.29% less latency than

the without caching and 67.4% less latency than the simple caching technique. The proposed approach has consumed 92.6% less energy than the without caching method and 82.7% less energy than the simple caching method.

The aim is to find the file popularity and top node selection using a machine learning technique. It is similar to the cold start method which initially randomly assigns the node and then using the previous data and learning patterns, the fog layer decides the top node and files. The introduction of machine learning and data logs will help in improving more energy and performance of the fog layer.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Muzammil Hussain Shahid: Conceptualization, Methodology, Writing - original draft. **Ahmad Raza Hameed:** Software, Validation, Formal analysis, Data curation, Writing - original draft. **Saif ul Islam:** Visualization, Writing - review & editing. **Hasan Ali Khattak:** Supervision, Project administration. **Ikram Ud Din:** Investigation, Writing - review & editing. **Joel J.P.C. Rodrigues:** Conceptualization, Resources, Supervision, Writing - review & editing.

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